
EXPERIENCE

- **Tunga** Kampala, Uganda
Backend Software Engineer (Django) Nov 2023 - Present
 - **Architected AI Talent Matching and Outreach Agent (Tunga-agent)**: Led end-to-end development of a multi-stage AI agent pipeline for developer matching and automated client outreach. Uses **RAG** with Pinecone vector search and OpenAI embeddings to semantically match developers to vacancies, generates personalized outreach emails via **GPT-4o-mini**, and automates follow-up sequences triggered by SendGrid engagement webhooks. Integrated Hunter.io for contact enrichment and Clay for lead ingestion. Supports hybrid LLM mode (OpenAI + local **Ollama**), reducing API costs by 90%. Deployed on **Kubernetes** with **Celery/Redis** background processing.
Core Technologies: Python, Django, OpenAI, Pinecone, Ollama, Celery, Redis, SendGrid, Kubernetes.
 - **Built AI-powered B2B Lead Identification System (Lead-ID)**: Designed and led development of a multi-signal B2B lead scoring engine to expand Tunga's offering from talent matching to outbound lead generation. The system monitors companies for buying signals — hiring spikes, vendor comparisons, funding events — across 15+ configurable signal types with recency decay weighting, and surfaces high-priority leads with AI-generated "why now" outreach narratives. Built a configuration dashboard enabling the marketing team to tune signal weights and thresholds without code changes. Processes **500+ companies per batch** with score delta tracking across runs.
Core Technologies: Python, Django, OpenAI, Celery, Redis, PostgreSQL, Clay.
 - **Developed an AI-powered resume parser**: Implemented a feature that automatically populates developer profiles during onboarding by intelligently parsing resumes. This improved the efficiency of the profile creation workflow and reduced manual entry for developers.
 - **Optimized platform performance and billing**: Diagnosed and resolved "N + 1 query" issues across several backend endpoints, leading to significant API performance improvements. Implemented multi-line and multi-developer invoicing, and expanded payment options to USD (previously EUR-only), reducing currency conversion fees for developers.
- **Sunbird AI** Kampala, Uganda
Software Engineer and Team Lead May 2020 - Oct 2023
 - **Leadership**: Provided technical leadership and mentorship to a team of developers and interns, ensuring the successful productization of ML models and effective management of MLOps workflows.
 - **Model Deployment**: Spearheaded the deployment of language translation models to an Amazon Sagemaker endpoint, facilitating seamless integration through an API for external users. Later on, led the migration from Sagemaker to GCP Vertex AI. These models power the Sunbird API.
 - **Language Translation API**: Developed and led the implementation of the Sunbird AI API, offering language technology services including translation, speech recognition, and text-to-speech in Ugandan local languages. Currently, the API is being utilised by Ugandan companies within their applications and was used in the IndabaX language technology hackathon.
 - **Language Translation Frontend**: Revamped the front end of the Ugandan languages translation app, transitioning it from vanilla Javascript to React, thus enabling faster feature development and eliminating a whole class of state-related bugs. This app played a pivotal role in the successful launch of Sunbird AI.
 - **Noise Monitoring**: Established the end-to-end infrastructure (including back end and front end) for the Environmental Noise Monitoring System in Kampala and Entebbe. Subsequently, orchestrated the migration of core components from a self-hosted setup to AWS IoT, significantly enhancing reliability and robustness. The Mayor of Entebbe relies on this system for monitoring and mitigating noise in the city.
- **Andela** Kampala, Uganda
Software Engineer Aug. 2018 – Apr. 2020
 - **Android development**: Built offline-functionality features for the Converge app (an internal meeting room management mobile app). Also improved how we wrote instrumented tests to mimic offline situations using the Test Butler library.
 - **Leadership**: Mentored new developers during their apprenticeship period by assigning them tasks, conducting code reviews and providing technical guidance.
- **Makerere University Data Science and Artificial Intelligence Lab** Kampala, Uganda
Software Engineer Intern Jun. 2017 – Sept. 2017
 - **Android development**: Developed a photo-annotation feature in the Android White-Fly count app which helped farmers to annotate pictures of white-fly infested cassava leaves.

EDUCATION

- **Makerere University** Kampala, Uganda
Bachelor of Science in Computer Science *Aug. 2015 - Apr. 2018*
- **Coursera Courses** online
key specializations
 - Machine Learning Specialization
 - Mathematics for Machine Learning
 - Deep Learning Specialization

PROJECTS

- **AcuGemma: On-Device AI Tutoring App**
 - **Developed an on-device AI tutoring application for primary school students:** Integrated the Gemma 3n 2B LLM directly onto an Android device using MediaPipe, which ensures user privacy, rapid response times, and full offline capability. The application features a dynamic prompting system that guides lessons, evaluates student responses, and provides context-aware feedback.
- **Uganda Coffee Traceability Dashboard: Geospatial Data Visualization & Verification Tool**
 - **Developed a dashboard to verify 97,000+ coffee plots against 2020 satellite baselines.:** Integrated 10m-resolution Google Earth Engine data into a GeoDjango and PostGIS backend, featuring a GraphQL API for real-time coordinate verification and a React-Leaflet interface that manages high-density spatial layers through district-level filtering and geometry optimization.

PROGRAMMING SKILLS

- **Languages:** Python, Javascript, Kotlin **Technologies:** Django, React, FastAPI, AWS, GCP, Celery/Redis, Pinecone, OpenAI API, Kubernetes